AI-LAB ASSIGNMENT- 7.2

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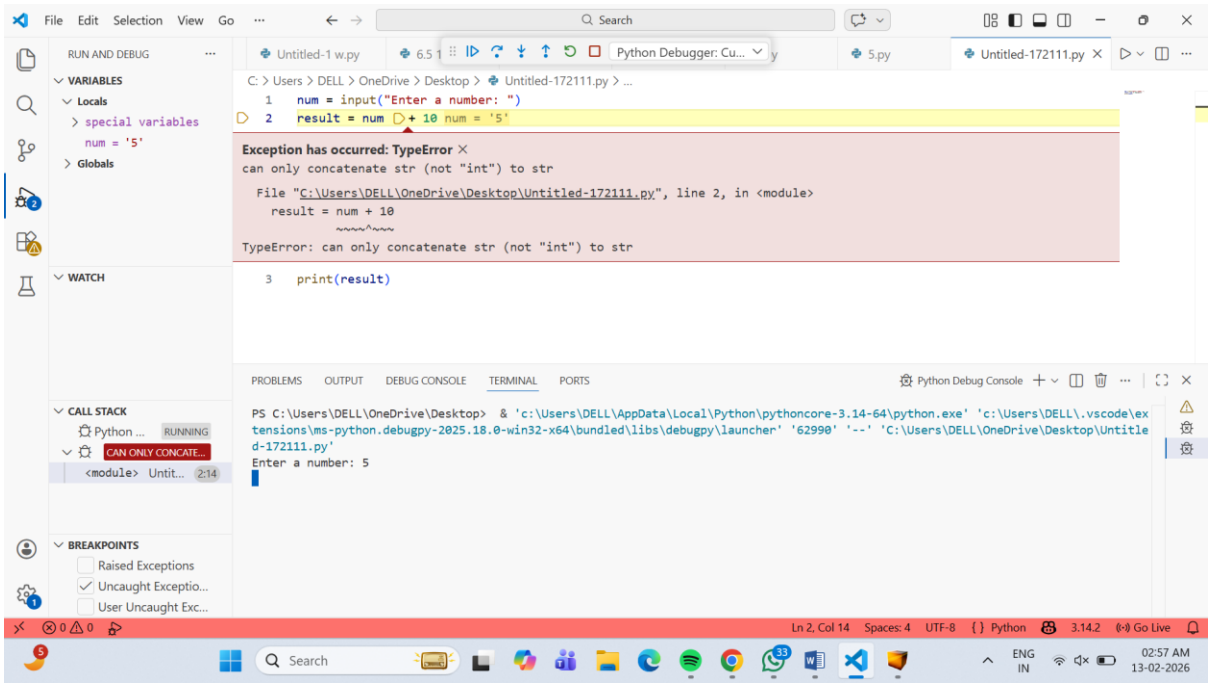
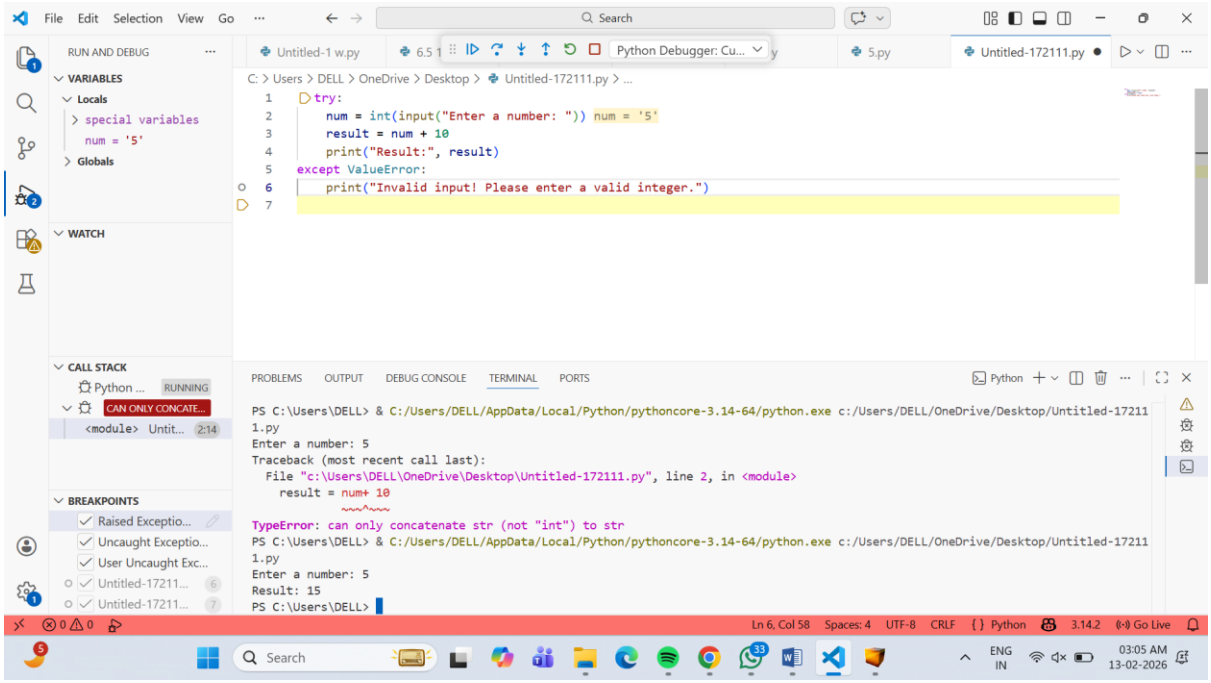
LAB 7: ERROR DEBUGGING WITH AI: SYSTEMATIC APPROACHES TO FINDING AND FIXING BUGS

TASK 1 – RUNTIME ERROR DUE TO INVALID INPUT TYPE

● A Python program accepts user input and performs arithmetic operations. However, the program throws a runtime error because the input is treated as a string instead of a numeric type.

CODE :





JUSTIFICATION:

- int(input()) converts the string input into an integer.
- Arithmetic operations can now be performed safely.
- The runtime error is eliminated.

TASK 2 – INCORRECT FUNCTION RETURN VALUE

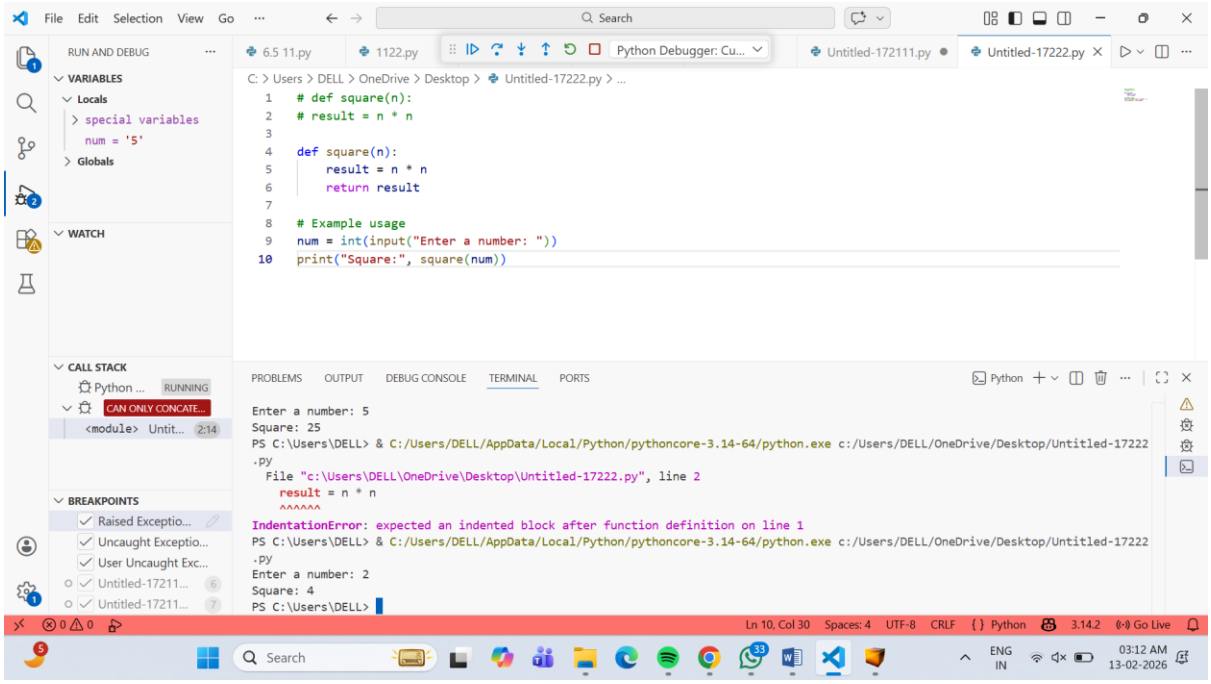
A function is designed to calculate the square of a number, but it does not return the computed result properly.

Example (Buggy Code):

def square(n):

result = n * n

CODE :



JUSTIFICATION:

- n * n correctly computes the square

. • return result ensures the value is passed back.

- The function now behaves as expected.

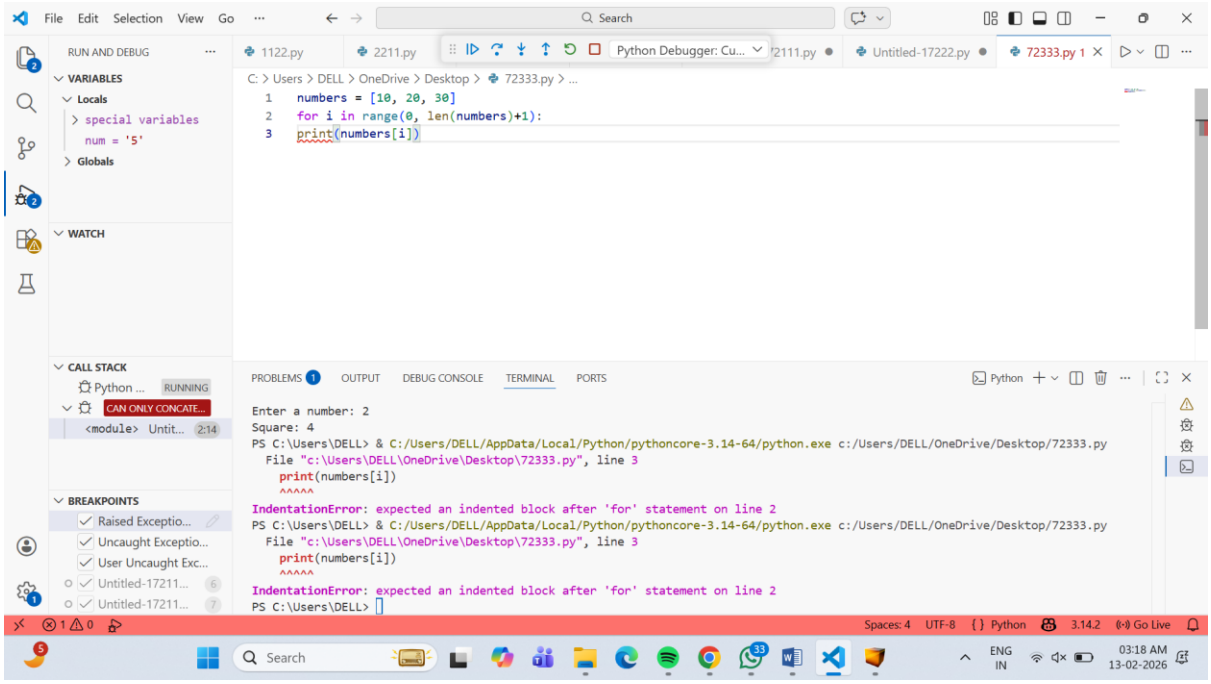
TASK 3 – INDEXERROR IN LIST TRAVERSAL

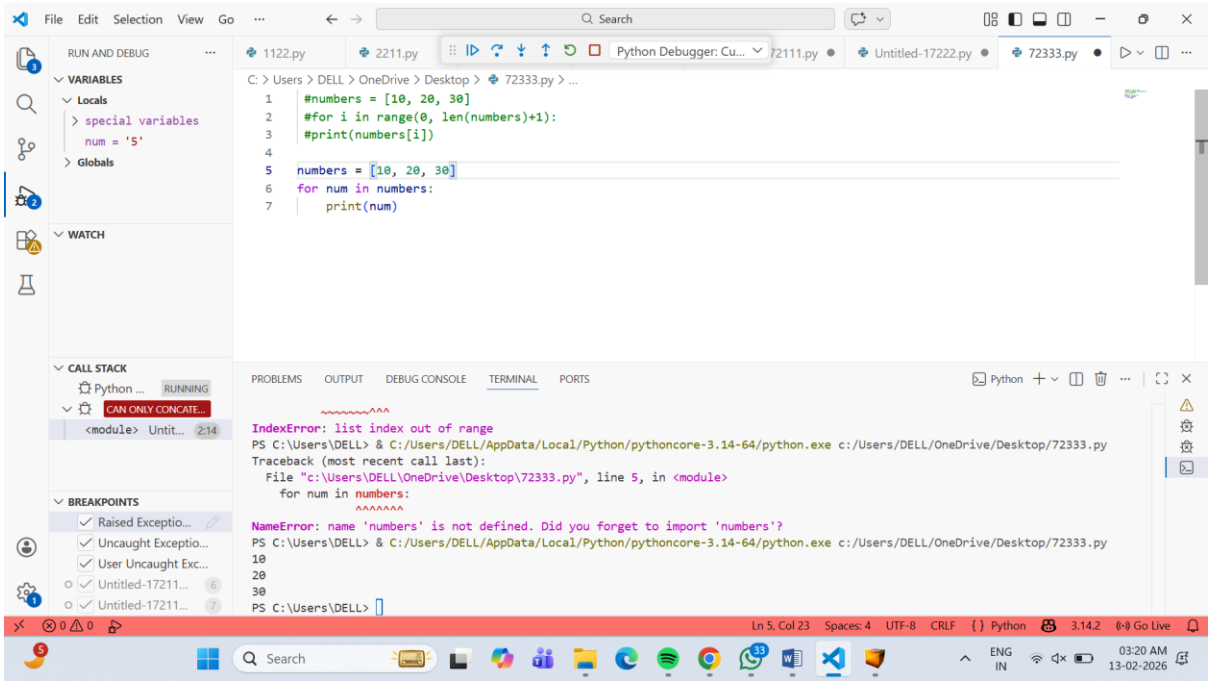
A Python program iterates over a list using incorrect index limits, causing an IndexError.

Example (Buggy Code):

```
numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
print(numbers[i])
```

CODE:





JUSTIFICATION:

- range (0, len(numbers)) prevents out-of-range access.
- The enhanced loop avoids index handling entirely.
- The Index Error is fully eliminated.

TASK 4 – UNINITIALIZED VARIABLE USAGE

A program uses a variable in a calculation before assigning it any value.

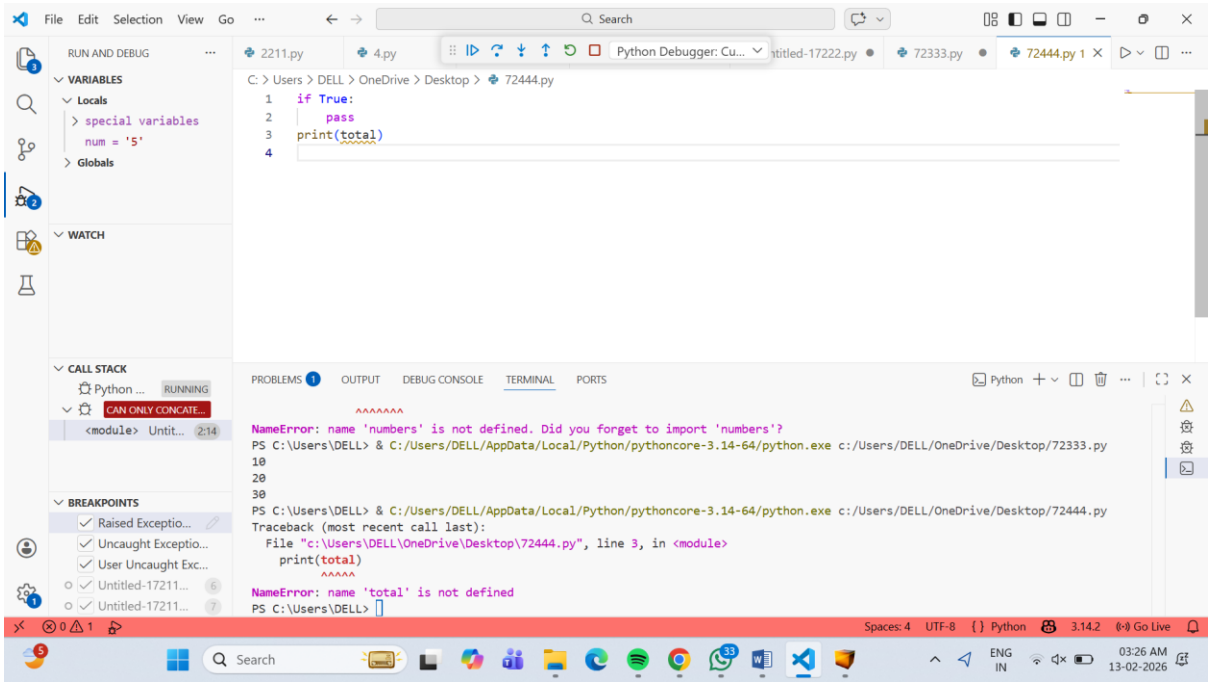
Example (Buggy Code):

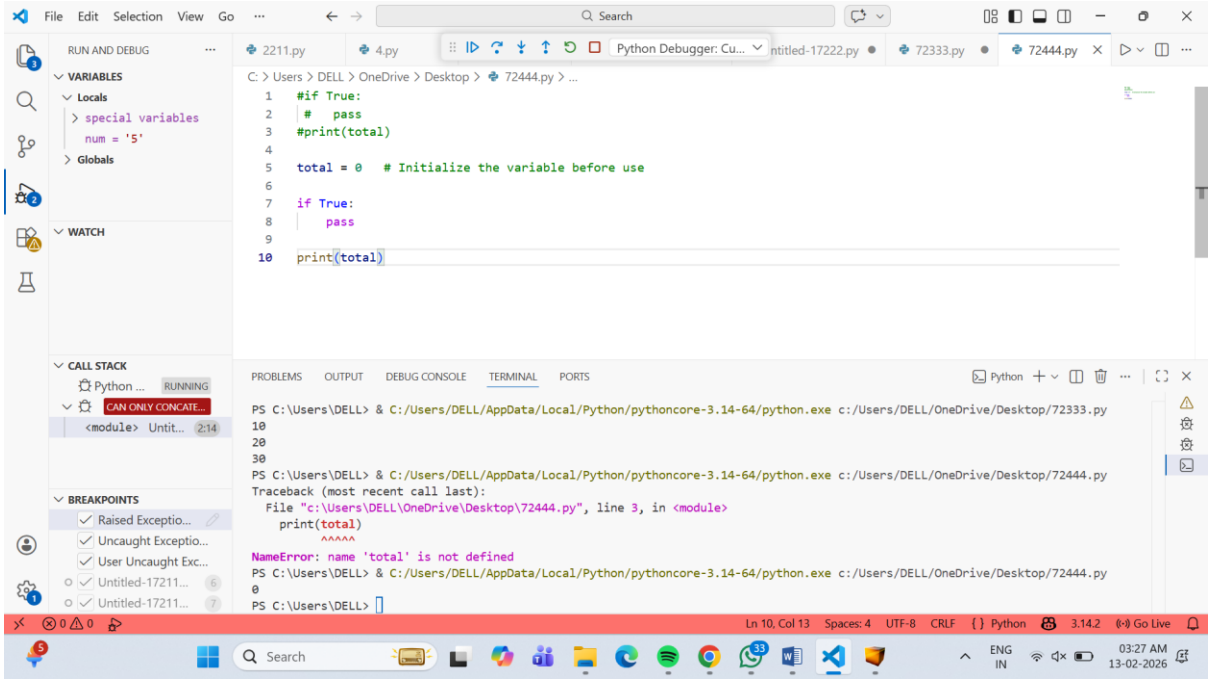
if True:

pass

print(total)

CODE :





JUSTIFICATION:

• Initializing total ensures it exists in memory

. • The program can now safely access and print the variable.

• The runtime error is eliminated.

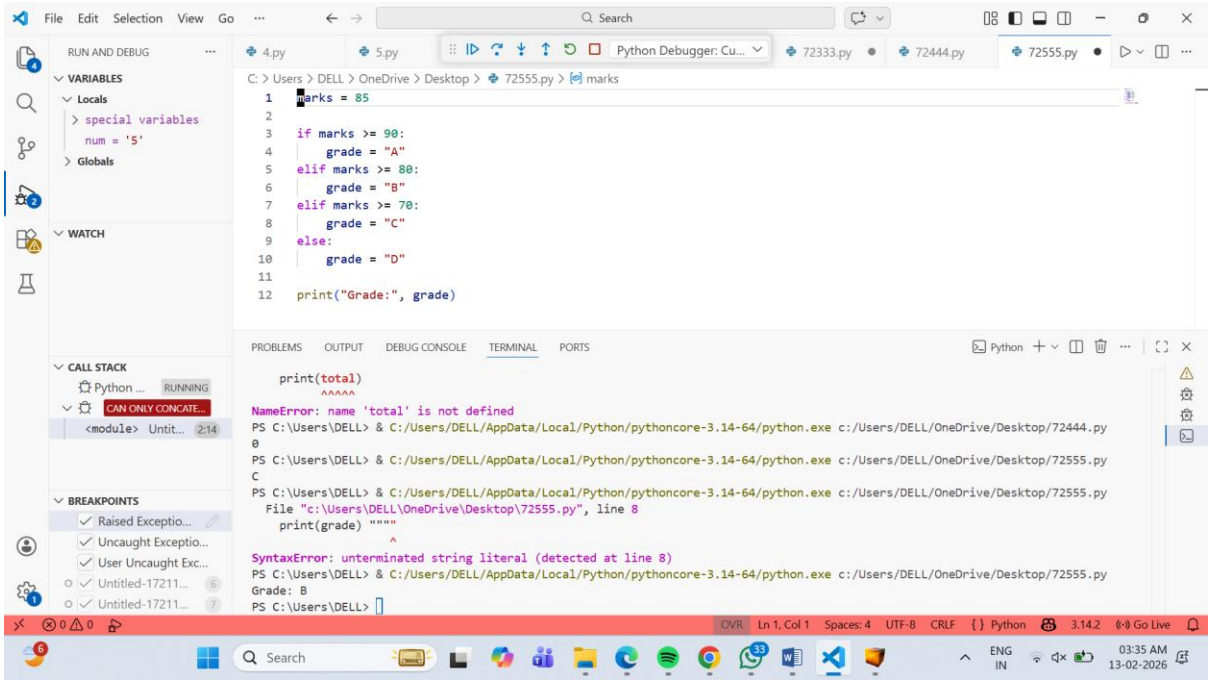
TASK 5 – LOGICAL ERROR IN STUDENT GRADING SYSTEM

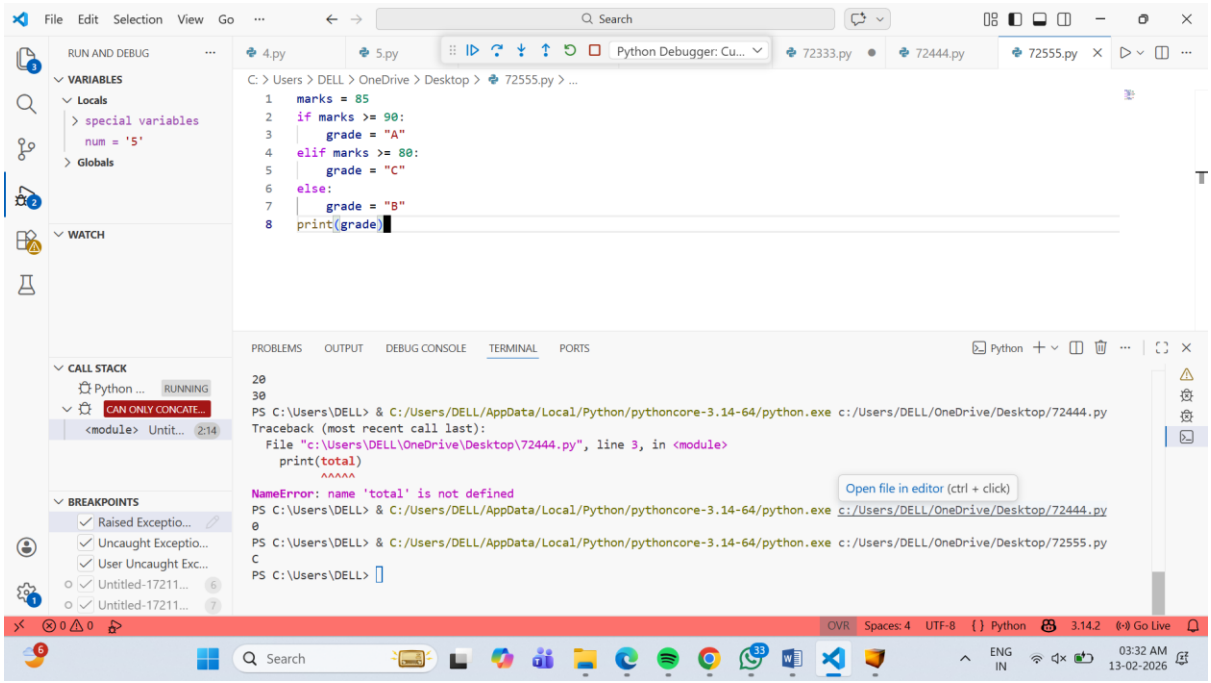
A grading program assigns incorrect grades due to improper conditional logic.

Example (Buggy Code):

```
marks = 85
if marks >= 90:
grade = "A"
elif marks >= 80:
grade = "C"
else:
grade = "B"
print(grade)
```

CODE:





JUSTIFICATION:

- Initializing total ensures it exists in memory.
- The program can now safely access and print the variable.
- The runtime error is eliminated.